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Decoupling Control Optimization Based on Analysis and Compensation of Back EMF Disturbance in High Precision Current Converter

JIAN WEI¹, BAOQUAN KOU¹, (Member, IEEE), LU ZHANG¹, (Member, IEEE),
HAILIN ZHANG², (Member, IEEE), AND DEXUE SONG¹

¹Department of Electrical Engineering, Harbin Institute of Technology, Harbin 150080, China

²Postdoctoral Station of Electrical Engineering, Chongqing University, Chongqing 400044, China

Corresponding author: Baoquan Kou (koubaq@hit.edu.cn)

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ABSTRACT In high performance applications, LC filter usually is added to power converter for optimizing the output current ripple characteristics of the system. However, the LCL link composed of LC filter and resistor-inductance load causes the resonance, which leads to the traditional closed-loop not appropriate to be applied directly. Consequently, the active damping compensation method based on harmonic current is considered to suppress the resonance, and the suppression effects of different damping parameters on the resonant peak are compared and analyzed. Moreover, back EMF treated as a nonlinear disturbance affects the accuracy of current. Therefore, the current harmonic suppression method based on feedforward control for resonant pole-based dual buck soft-switching converter (RPDBSC) is presented, and the current suppression method of feedforward control is analyzed by the decoupling control model of bias current. This paper focuses on the improvement of the closed-loop control strategy for high precision current converters, and the effectiveness of the proposed method is verified by simulation and experiment.

INDEX TERMS Active damping compensation, back EMF, nonlinear disturbance, feedforward control, dual buck soft-switching converter.

I. INTRODUCTION

For high precision motion applications, the design of current loop is important, such as precision motion platform for semiconductor manufacturing equipment [1], [2]. As the innermost loop of system, the development of current loop can improve the rapidity of the system, and suppress the influence of nonlinear factor. The current converter is applied to drive voice coil motor or planar motor in motion system [3], [4]. The precise control of current is the basis of precise torque control, and the accuracy of motor torque control determines the positioning accuracy of the motion system. Therefore, in the high-precision motor motion platform, the current harmonic suppression is indispensable [5], [6]. Due to no blanking time, dual buck topology is suitable for high

precision application [7]. In order to improve the response speed of the converter, the soft-switching technology and dual buck topology are combined to make the converter operate in high frequency. The bias current is injected into the inductor by decoupling control, which ensures the inductor operate in continuous current conduction mode (CCM) to eliminate the zero crossing distortion [8].

The output current of PWM power converter contains many high-order harmonics, and LC filter is usually applied to suppress high-frequency harmonics to achieve better current quality. However, LC filter and resistor-inductance load constitute the LCL filter on the output side. A resonance peak occurs at the resonance frequency of LCL filter, and the phase degree changes -180° . The output current of the converter oscillates due to the resonant peak, which affects the stability of system [9]. For realizing a stable system, the resonant peak needs to be suppressed, and the corresponding

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methods can be classified into passive damping and active damping. The passive damping method has the advantages of simple structure and insensitivity to system state [10]. The typical passive damping strategy is resistance and filter inductor (capacitor) in series or parallel. The resistance and inductor in series influence low-frequency gain of the filter. In addition, the passive damping method by adding resistance has relatively large losses [11]. In order to avoid the above disadvantages of passive damping method, virtual resistance can be adopted to replace the actual resistance, which is active damping method applied to suppress the resonant peak of LCL filter [12], [13]. The active damping method based on capacitance current feedback can effectively restrain the resonance peak of LCL filter [14], [15], [16]. In addition to the active damping method, state feedback control [17], [18], predictive control [19], [20], and other control strategies [21], [22] can also restrain the resonance peak and improve the dynamic response performance of the system [10]. However, the controller design and implementation of these control strategies are complex, and they have some impacts on the accuracy of the current.

As a nonlinear disturbance, back EMF affects the current accuracy of the converter. In [23], a neural-network method is proposed for current harmonics suppression with nonsinusoidal back-EMF of permanent magnet synchronous motors. By generating compensation voltage in forward channel, back EMF is compensated and current harmonics are suppressed. Based on the control strategy and the characteristics of non-ideal back EMF, the extended state observer is established to estimate the back EMF in [24]. In order to solve the high-speed position sensing issues such as digital conversion and inductance cross coupling, a discrete back EMF voltage estimation method is proposed [25]. In [26], a control strategy based on eliminating back EMF harmonics and insensitive to motor parameters is developed for the improvement of motor position estimation accuracy, and the strategy can compensate the influence of asymmetric and non-sinusoidal back EMF. In order to suppress interference during current control, a control method with back EMF compensation is proposed, and it has the similar effect compared with the variable gain PI control on current ripple and current response [27]. In [28], a method of generalized sliding discrete Fourier transform (GSDFT) for extracting harmonic components is analyzed, and the corresponding compensation voltage is calculated to compensate the harmonic components according to the motor mathematical model. After nonlinear compensation, the total harmonic distortion of current is reduced. When the motor operates at low speed, the effect of the proposed method is limited. In order to ensure the medium and high speed sensorless control performance of fault-tolerant permanent-magnet synchronous motor in the case of open and short circuit fault, a robust observer is proposed to estimate the back EMF of two normal phases without considering parameter variations, external disturbances and phase faults [29]. In [30], a control strategy including PI regulator, stator resistance voltage drop and back EMF compensation

are proposed to suppress the influence of back EMF on current.

Most researches of active damping method focus on grid connected converter. There are few researches in the field of motor precision drive, especially the damping compensation based on high precision converter is still missing. Furthermore, the compensation of back EMF has achieved some results, and less attention is paid to the achievable current accuracy in the published literature. The load of high-precision converter is usually considered as resistor-inductance load, so it is necessary to research the influence of back EMF on current accuracy and its suppression method. Based on the above analysis, this paper focuses on the closed-loop control strategy of high-precision power converter. The main contents of this paper are organized as follows: In Section II, the operation principle of the converter is introduced, and the decoupling control is analyzed. In Section III, the active damping of harmonic current feedback based on decoupling control model is presented, and the damping effects of different feedback coefficients are compared. In Section IV, the influence of back EMF on the output current accuracy of high precision converter and the suppression method of feedforward control are analyzed. In Section V, the experimental results of the high precision current converter prototype are presented.

II. OPERATING PRINCIPLE

A. OPERATION OF THE HIGH-PRECISION CURRENT CONVERTER

The resonant pole-based dual buck soft-switching topology is shown in Fig. 1, which consists of two dual buck legs with the load represented by R , L and back EMF, the auxiliary resonant circuit, and the LC filter. Due to the symmetry of the dual buck full bridge circuit, the soft-switching resonant cells are also symmetrical, and the four resonant cells have the similar structure. The auxiliary branch composed of S_{11} , L_{r1} and D_{s1} corresponds to the bridge leg composed of S_1 and D_1 , and other bridge legs are similar. The proper operation can make the resonant circuit operate according to the resonant law, which provides zero voltage switching (ZVS) conditions for the main switches S_1 - S_4 . After the discharge of the parallel capacitors C_{s1} - C_{s4} (parasitic capacitors are included) of the main switches, the reverse voltage is provided for the resonant inductors to make the resonant current return to

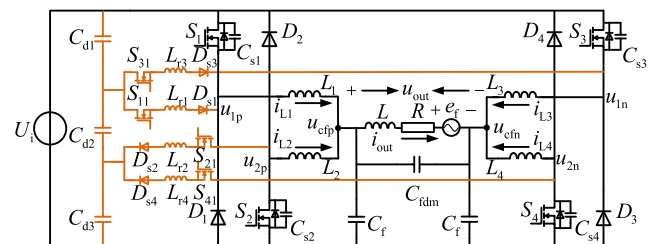


FIGURE 1. The topology of resonant pole-based dual buck soft-switching converter.

zero. Therefore, the auxiliary switches S_{11} - S_{41} can achieve zero current switching (ZCS) conditions. The time sequence transition of soft-switching converter is shown in Fig. 2. The input voltage is divided by the three capacitors for applying in auxiliary circuit. Due to the unidirectional charge or discharge, the additional voltage balance strategy is needed to maintain the balance of capacitance voltage.

B. CONTROL METHOD OF THE CURRENT CONVERTER

The operation of the auxiliary resonant circuit depends on the state of the main circuit and the auxiliary circuit is compatible with the operation of the main circuit. Based on the average voltage of switching nodes, a state-space average model of soft-switching topology is established. The state equations (1)–(3), as shown at the bottom of the page.

In the above formulas, R_f and L_f are the resistance and inductance of inductance L_1 , L_2 , L_3 and L_4 . A, B and C are coefficient matrices.

$$\begin{bmatrix} i_{out} \\ i_{biasp} \\ i_{biasn} \\ i_{fdm} \\ i_{fcm} \end{bmatrix} = H \begin{bmatrix} i_{out} \\ i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & -1/2 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & -1/2 \\ -1 & 1/2 & 1/2 & -1/2 & -1/2 \\ 0 & 1 & 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} i_{out} \\ i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \end{bmatrix} \quad (4)$$

$$\begin{bmatrix} u_{dm} \\ u_{cm} \\ u_{biasp} \\ u_{biasn} \end{bmatrix} = Gu = \begin{bmatrix} 1/2 & 1/2 & -1/2 & -1/2 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} u_{1p} \\ u_{2p} \\ u_{1n} \\ u_{2n} \end{bmatrix} \quad (5)$$

By substituting the decoupling transformation into (1) and (2), the state equation can be transformed as (6) and (7), shown at the bottom of the next page.

$$\begin{aligned} \dot{x} &= Ax + Bu + Z \\ &= \begin{bmatrix} -R/L & 0 & 0 & 0 & 0 & 1/L & -1/L \\ 0 & -R_f/L_f & 0 & 0 & 0 & -1/L_f & 0 \\ 0 & 0 & -R_f/L_f & 0 & 0 & -1/L_f & 0 \\ 0 & 0 & 0 & -R_f/L_f & 0 & 0 & -1/L_f \\ 0 & 0 & 0 & 0 & -R_f/L_f & 0 & -1/L_f \\ -a_1 & a_2 & a_2 & a_3 & a_3 & 0 & 0 \\ a_1 & a_3 & a_3 & a_2 & a_2 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} i_{out} \\ i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \\ u_{cfp} \\ u_{cfn} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 1/L_f & 0 & 0 & 0 & 0 \\ 0 & 1/L_f & 0 & 0 & 0 \\ 0 & 0 & 1/L_f & 0 & 0 \\ 0 & 0 & 0 & 1/L_f & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_{1p} \\ u_{2p} \\ u_{1n} \\ u_{2n} \end{bmatrix} + \begin{bmatrix} -e_f/L \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (1) \end{aligned}$$

$$y = Cx = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} i_{out} \\ i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \\ u_{cfp} \\ u_{cfn} \end{bmatrix} \quad (2)$$

$$\begin{cases} a_1 = \frac{1}{C_f + 2C_{fdm}} \\ a_2 = \frac{1}{2C_f} + \frac{1}{2(C_f + 2C_{fdm})} \\ a_3 = \frac{1}{2C_f} - \frac{1}{2(C_f + 2C_{fdm})} \end{cases} \quad (3)$$

After decoupling transformation, the decoupling input-output equation can be derived as follows [31]. From (1)-(8), it can be seen that the decoupling control of bias current and output current can be realized by decoupling transformation. The bias current and output current can be controlled independently.

$$\begin{aligned}
 Y'(s) &= C'(sI-A)^{-1}(B'u' + Z(s)) \\
 &= \begin{bmatrix} p_{out} & 0 & 0 & 0 \\ 0 & 0 & p_{bias} & 0 \\ 0 & 0 & 0 & p_{bias} \\ p_{dm} & 0 & 0 & 0 \\ 0 & p_{cm} & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_{dm} \\ u_{cm} \\ u_{biasp} \\ u_{biasn} \end{bmatrix} \\
 &+ \begin{bmatrix} q_{out} \\ 0 \\ 0 \\ q_{dm} \\ 0 \end{bmatrix} \cdot e_f \quad (8)
 \end{aligned}$$

From the Fig. 3, the output current and bias current are decoupled and they have respective control loops. By injecting bias current into the filter inductor to make the inductor operate in CCM, and the zero crossing distortion of the output current can be eliminated. The active damping compensation

module k_{dm} and feed forward control module $G_c(s)$ are included in the control block diagram.

III. ACTIVE DAMPING COMPENSATION

According to the structure diagram in Fig. 4, when there is no harmonic current feedback, the transfer functions ($G_T(s)$) of output current channel is shown as (10), at the bottom of the next page.

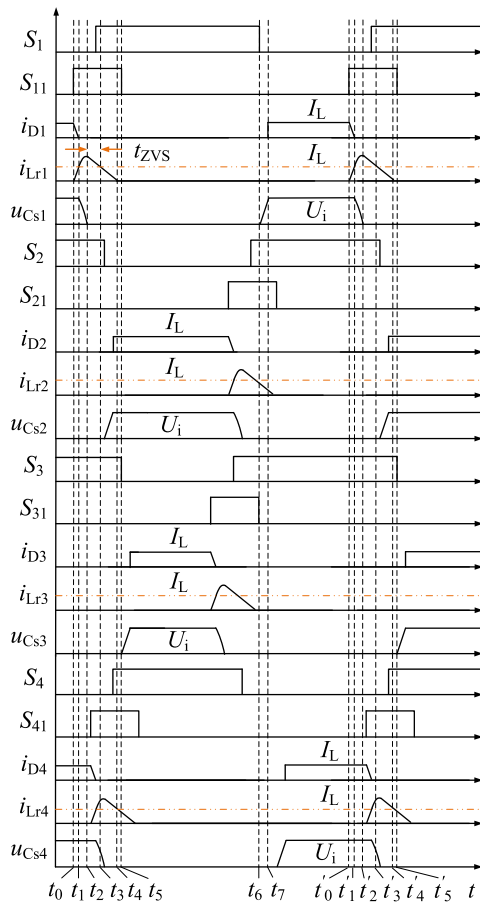
The current regulator $C_{out}(s)$ is a PI controller, and the expression of the controller is shown as (9), at the bottom of the next page.

Fig. 6 shows the closed-loop bode diagram of the system, and the corresponding root locus is shown in Fig. 5. Because LC filter and resistor-inductor load constitute LCL filter link, there is a resonant peak at the resonant frequency based on the frequency response characteristics of the system. The resonance peak makes the system state in critical stable, which can be verified by the root locus in Fig. 5. In addition, the phase frequency characteristic shown in Fig. 6 is reversed by 180° . Therefore, it is necessary to introduce appropriate damping to calibrate the frequency characteristics, suppress the resonance peak and improve the stability of the system.

After introducing harmonic current feedback, the control block diagram is shown as Fig. 7. Based on the structure

$$\begin{aligned}
 \dot{x} &= Ax + B'u' + Z \\
 &= \begin{bmatrix} -R/L & 0 & 0 & 0 & 0 & 1/L & -1/L \\ 0 & -R_f/L_f & 0 & 0 & 0 & -1/L_f & 0 \\ 0 & 0 & -R_f/L_f & 0 & 0 & -1/L_f & 0 \\ 0 & 0 & 0 & -R_f/L_f & 0 & 0 & -1/L_f \\ 0 & 0 & 0 & 0 & -R_f/L_f & 0 & -1/L_f \\ -a_1 & a_2 & a_2 & a_3 & a_3 & 0 & 0 \\ a_1 & a_3 & a_3 & a_2 & a_2 & 0 & 0 \end{bmatrix} \\
 &\cdot \begin{bmatrix} i_{out} \\ i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \\ u_{cfp} \\ u_{cfn} \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1/(2L_f) & 1/L_f & 1/(2L_f) & 0 \\ 1/(2L_f) & 1/L_f & -1/(2L_f) & 0 \\ -1/(2L_f) & 1/L_f & 0 & 1/(2L_f) \\ -1/(2L_f) & 1/L_f & 0 & -1/(2L_f) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} u_{dm} \\ u_{cm} \\ u_{dm} \\ u_{dm} \end{bmatrix} + \begin{bmatrix} -e_f/L \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (6)
 \end{aligned}$$

$$\begin{aligned}
 y' &= \begin{bmatrix} i_{out} \\ i_{biasp} \\ i_{biasn} \\ i_{fdm} \\ i_{fcm} \end{bmatrix} = C'x \\
 &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & -1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & -1/2 & 0 & 0 \\ -1 & 1/2 & 1/2 & -1/2 & -1/2 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} i_{out} \\ i_{L1} \\ i_{L2} \\ i_{L3} \\ i_{L4} \\ u_{cfp} \\ u_{cfn} \end{bmatrix} \quad (7)
 \end{aligned}$$



The diagram shows a closed-loop control system. The reference input $R_{iout}(s)$ is compared with the feedback signal at a summing junction. The error signal is processed by the controller $C_{out}(s)$. The output of the controller is the input to the plant, which is enclosed in a dashed box labeled $p_{out}(s)$. Inside this box, the signal passes through a PWM block K_{pwm} , a filter $G_f(s)$, and a delay $G_d(s)$. The output of the plant is $C_{iout}(s)$, which is then fed back through a gain block k to the summing junction.

According to frequency characteristic of the system shown in Fig. 9, the introduction of harmonic current feedback can suppress the resonance peak effectively. Compared with the closed-loop control without damping, the active damping makes the root of the closed-loop characteristic equation move to the left half plane of the complex frequency domain (S domain) shown in Fig. 8. Consequently, active damping

Based on different feedback coefficients k_{dm} , the influence of applying harmonic current feedback on the root locus is analyzed. Fig. 9 and Fig. 11 show the closed-loop pole

$$G_T(s) = \frac{C_{out}(s)p_{out}(s)}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)} = \frac{k_p s + k_i}{\alpha L L_f s^4 + (\alpha \beta + k_{dm} \alpha L) s^3 + (L + L_f + \alpha R R_f + \alpha k_{dm} R) s^2 + (R + R_f + k k_n) s + k k_i} \quad (11)$$

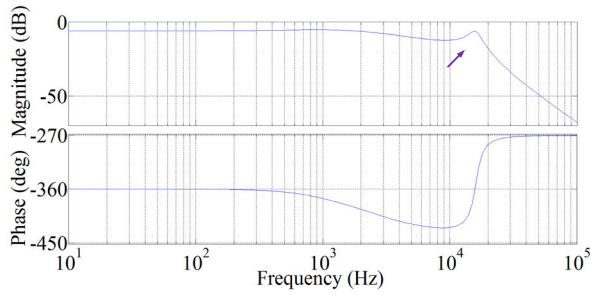


FIGURE 6. $k_{dm} = 0$, bode diagram of the system.

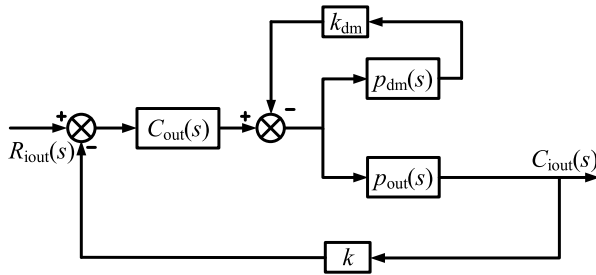


FIGURE 7. Structure diagram of harmonic current feedback.

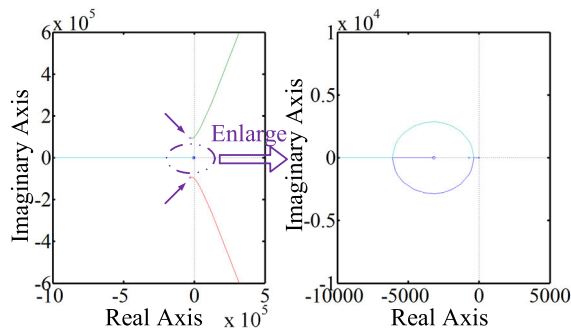


FIGURE 8. $k_{dm} = 20$, root locus of the system.

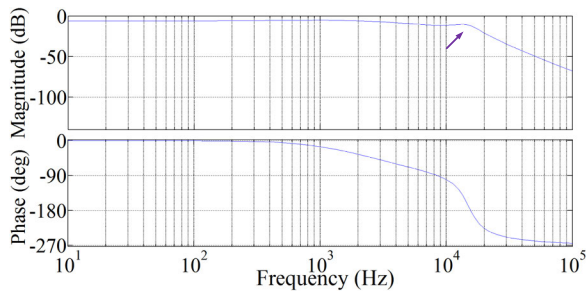


FIGURE 9. $k_{dm} = 20$, bode diagram of the system.

diagram of the system under the different harmonic current feedback coefficients $k_{dm} = 20$ and $k_{dm} = 50$ respectively. Accordingly, Fig. 8 and Fig. 10 show the root locus of the system corresponding to $k_{dm} = 20$ and $k_{dm} = 50$ respectively. Based on the analysis of the figure above, it can be derived that increasing the feedback coefficient can weaken the resonance peak more effectively. With the increase of the

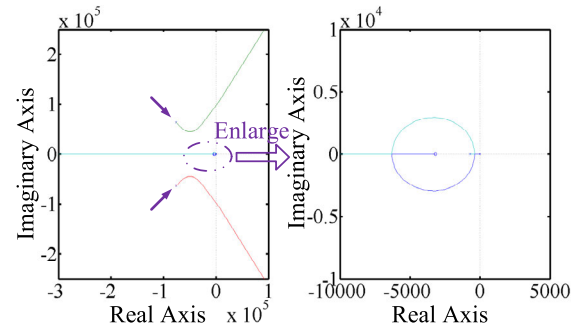


FIGURE 10. $k_{dm} = 50$, root locus of the system.

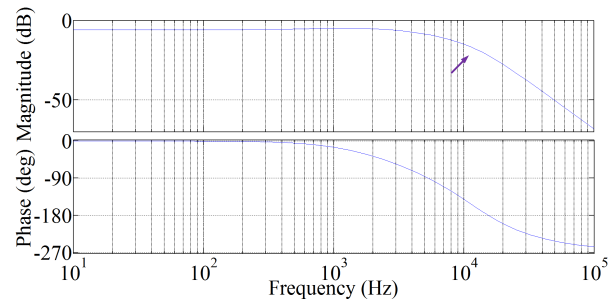


FIGURE 11. $k_{dm} = 50$, bode diagram of the system.

feedback coefficient, the characteristic root moves to the left plane, and the system is more stable. However, it should be pointed out that excessive damping affects the phase margin of the system, so it is necessary to choose a reasonable feedback coefficient. In the closed-loop bode diagram shown as Fig. 11, the closed-loop bandwidth of the system is more than 5 kHz, and the phase margin is greater than 60° . The resonant peak is effectively suppressed, which can meet the design requirements of the system.

IV. ANALYSIS AND COMPENSATION OF BACK EMF DISTURBANCE

In feedback control system, the input of the controller is determined by deviation. Under the influence of interference, the output tracking value deviates from the reference value, and the controller produces the command by comparing offset and reference. If the disturbance is applied continuously, the output of the system always follows the disturbance, which inevitably results in steady-state tracking error. Feed-forward control is an open-loop control that compensates error according to disturbance. In theory, feedforward control can completely eliminate the deviation caused by disturbance.

Based on the feedforward principle, feedforward control is applied to improve the control performance of current converter. Feedforward control does not depend on the output deviation to generate the control effect. When the external disturbance affects the system, the feedforward channel calculates the corresponding control command. It can effectively suppress the influence of external interference, reduce the

steady-state tracking error and improve the control accuracy of the system.

$$\frac{q_{out}(s)(1 + k_{dm}p_{dm}(s)) - k_{dm}q_{dm}(s)p_{out}(s)}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)}e_f \quad (12)$$

The influence of back EMF on the system is shown in Fig. 12. After closed-loop suppression, the interference caused by back EMF on the output current is derived as (12). The ability of suppressing the current disturbance caused by back EMF is related to the open-loop gain of the system. The larger the open-loop gain is, the stronger the disturbance suppression ability is. However, excessive gain affects the stability of the system.

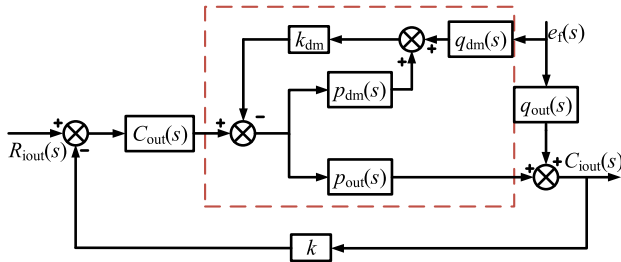


FIGURE 12. Influence of the back electromotive force (EMF) on the system.

Considering the back EMF, the transfer function of the output current channel are derived as shown in (13)–(14), at the bottom of the next page, which contains the components related to back EMF. The disturbance caused by back EMF affects the accuracy of the output current, and feedforward control needs to be introduced to compensate the disturbance of back EMF.

The transfer function of channel from input to output is shown in Fig. 13, which includes model of control object and feedforward control part.

According to the invariance principle, the disturbance error of back EMF to output current can be compensated to zero, so the feedforward controller can be derived as (15), shown at the bottom of the next page. In order to realize the feedforward control algorithm more appropriate, some higher-order components of the feedforward controller can be ignored.

The structure of feed-forward control is shown in Fig. 13, and the control structure consisted of feed-forward control and feedback control is adopted. The feed-forward control is used to compensate the influence of back EMF, and the dynamic performance of the system can be guaranteed. Fig. 14 shows the compensation effect of feed-forward control on back EMF. According to Fig. 14, it can be derived that feed-forward control can suppress the disturbance of back EMF to output current and reduces the current harmonic.

Compared with the condition of the EMF with harmonics, THD of output current is smaller under the condition of back EMF without harmonics. Combined with Fig. 15 and Fig. 16, it can be derived that the harmonic of back EMF leads to more current harmonic. The THD of the output current are equal to 0.02% and 0.07% under the condition of back EMF

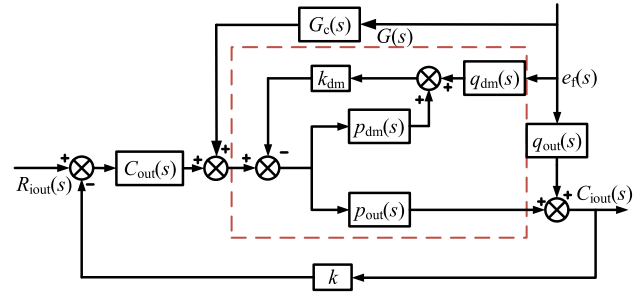


FIGURE 13. Structure diagram of feedforward control of the system.

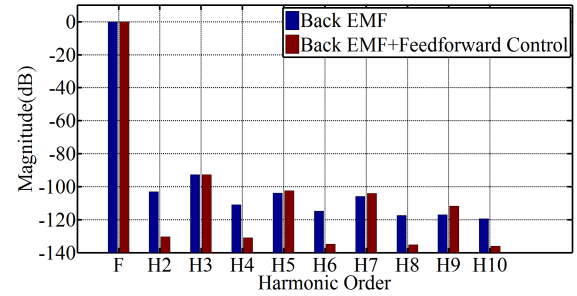


FIGURE 14. Influence of back EMF suppression by feedforward control on output current accuracy.

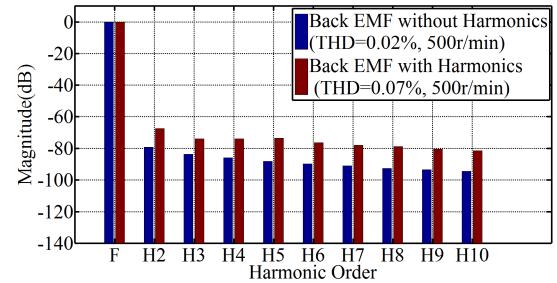


FIGURE 15. Influence of back EMF harmonic content (500r/min) on output current accuracy.

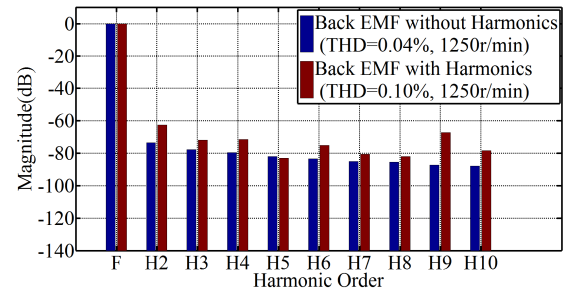


FIGURE 16. Influence of back EMF harmonic content (1250r/min) on output current accuracy.

with and without harmonic respectively when the speed of the load motor is 500 r/min. Similarly, when the speed of load motor is 1250 r/min, THD of output current are equal to 0.04% and 0.1%. The cogging torque and ripple torque of load motor cause the motor speed fluctuation, which increase

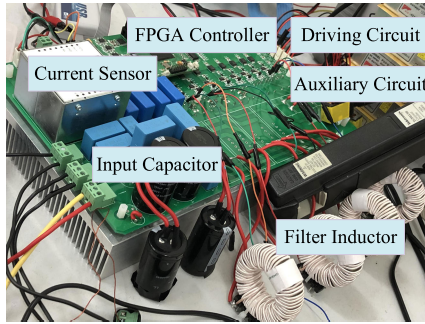


FIGURE 17. RPDBSC prototype.

TABLE 1. Circuit parameters of RPDBSC prototype.

Parameter	Value
DC-link Voltage U_i	500V
Switching Frequency f_s	100kHz
Filter inductor L_f	320μH
Filter capacitor C_f , C_{dm}	0.22μF
Resonant Inductance L_{r1} - L_{r4}	2.2μH
Resonant Capacitance C_{s1} - C_{s4}	1nF
MOSFETs S_1 - S_4 , S_{11} - S_{41}	IMW120R030M1H
DIODEs D_1 - D_4 , D_{s1} - D_{s4}	IDWD20G120C5XKSA1
Maximum Output Current	12.5A

the harmonic of the back EMF resulting in more current harmonics. In other words, when the motor with small cogging torque and ripple torque or the ironless core motor with low

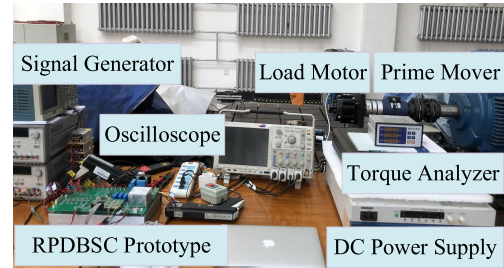


FIGURE 18. System experimental platform.

harmonic content of back EMF is used as load motor, the output current of RPDBSC realizes less harmonic in the same condition.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The RPDBSC was designed to operate at 500 V bus voltage and 100 kHz switching frequency as shown in Fig. 17. The control algorithm of the converter was built and implemented by applying a control unit based on digital processor (FPGA), and the components for the converters are given in Table 1. The experimental platform of the system is shown in Fig. 18. The large torque motor is used as the original motor to drive the load motor rotation, which makes the load motor operate in the stable speed (stable back EMF). Therefore, the influence of back EMF on the accuracy of output current and the effectiveness of closed-loop control algorithm are analyzed. In order to measure the current accuracy and waveform of RPDBSC, the reference signal is provided by

$$C_{iout}(s) = \frac{C_{out}(s)p_{out}(s)R_{iout}(s)}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)} + \frac{G_c(s)p_{out}(s)e_f}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)} + \frac{q_{out}(s)(1 + k_{dm}p_{dm}(s)) - k_{dm}q_{dm}(s)p_{out}(s)}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)}e_f \quad (13)$$

$$C_{iout}(s) = \frac{C_{out}(s)p_{out}(s)R_{iout}(s)}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)} + \frac{q_{out}(s)(1 + k_{dm}p_{dm}(s)) - k_{dm}q_{dm}(s)p_{out}(s)}{1 + k_{dm}p_{dm}(s) + kC_{out}(s)p_{out}(s)}e_f$$

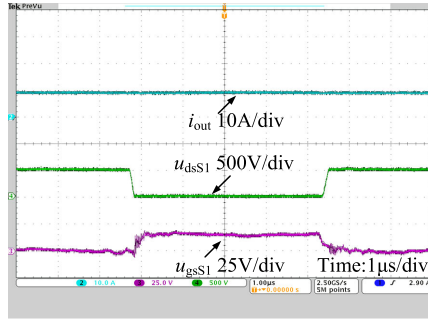
$$= \frac{k_p s + k_i}{\alpha L L_f s^4 + (\alpha \beta + k_{dm} \alpha L) s^3 + (L + L_f + \alpha R R_f + \alpha k_{dm} R) s^2 + (R + R_f + k k_p) s + k k_i} \cdot \frac{k_{dm}(\alpha L_f s^4 + \alpha R_f s^3)}{(\alpha L L_f s^3 + \alpha \beta s^2 + (L + L_f + \alpha R R_f) s + R + R_f)} e_f$$

$$= \frac{\left(\frac{k_{dm}(\alpha L s^2 + \alpha R s)}{\alpha L L_f s^3 + \alpha \beta s^2 + (L + L_f + \alpha R R_f) s + R + R_f} + 1 \right) (\alpha L_f s^4 + \alpha R_f s^3 + s^2) e_f}{k_{dm}(\alpha L s^4 + \alpha R s^3) + k(k_p s^2 + k_i s) + s^2(\alpha L L_f s^3 + \alpha \beta s^2 + (L + L_f + \alpha R R_f) s + R + R_f)} \quad (14)$$

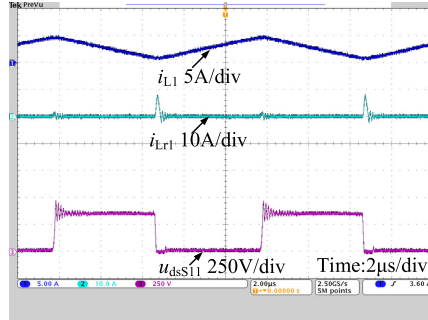
$$G_c(s) = k_{dm}q_{dm}(s) - \frac{(1 + k_{dm}p_{dm}(s))q_{out}(s)}{p_{out}(s)}$$

$$= \left(\frac{k_{dm}(\alpha L s^2 + \alpha R s)}{\alpha L L_f s^3 + \alpha \beta s^2 + (L + L_f + \alpha R R_f) s + R + R_f} + 1 \right) (\alpha L_f s^2 + \alpha R_f s + 1)$$

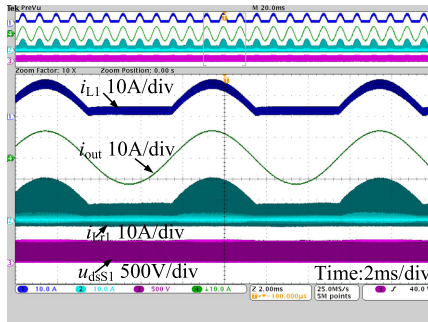
$$+ \frac{k_{dm}(\alpha L_f s^2 + \alpha R_f s)}{\alpha L L_f s^3 + \alpha \beta s^2 + (L + L_f + \alpha R R_f) s + R + R_f} \quad (15)$$



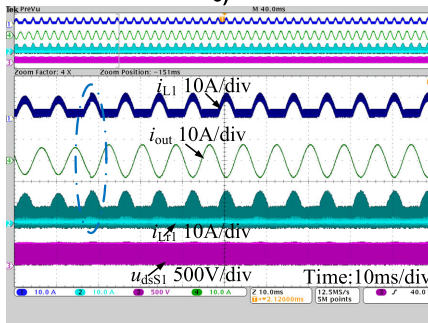
a)



b)



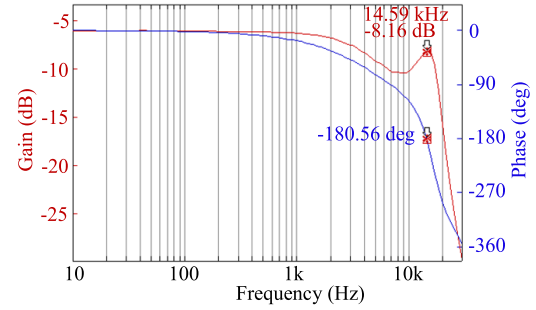
c)



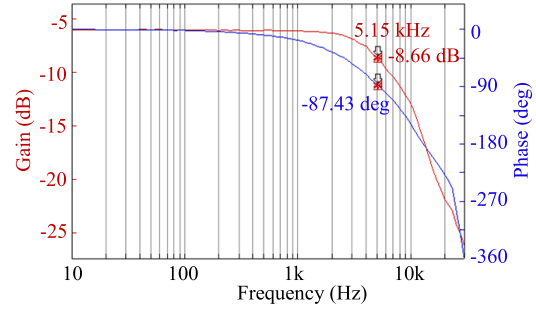
d)

FIGURE 19. Main waveform of RPDBSC topology. (a) Waveform of main switch (S_1). (b) Waveform of auxiliary switch (S_{11}). (c) Steady-state waveform. (d) Transient waveform.

the signal generator and the input voltage is provided by DC voltage source. The output current is measured and analyzed by oscilloscope and audio analyzer Apx515, and the results of current waveforms and current harmonic analysis are obtained.



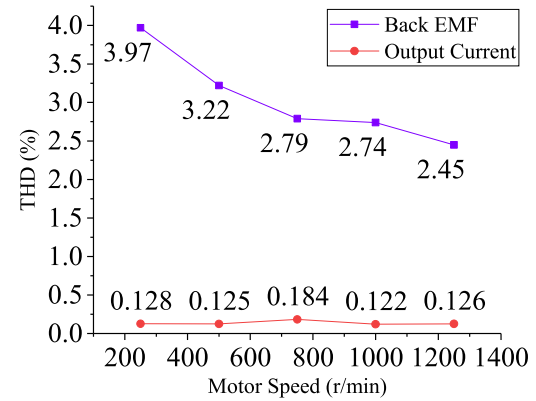
a)



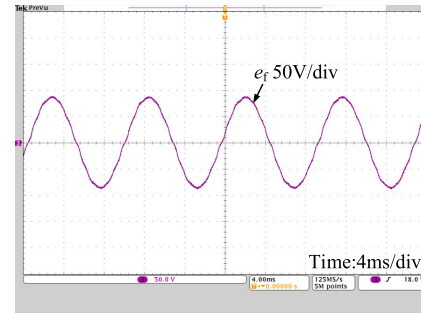
b)

FIGURE 20. Frequency characteristics of RPDBSC output current.

a) Damping coefficient $k_{dm} = 20$. b) Damping coefficient $k_{dm} = 50$.



a)



b)

FIGURE 21. Harmonic analysis of back EMF and output current. a) THD of EMF and load current. b) Waveform of e_f (1250r/min).

Fig. 19 shows the main waveforms of RPDBSC, including the output current, the drain source voltage of the main switch and the inductance current of the auxiliary resonant

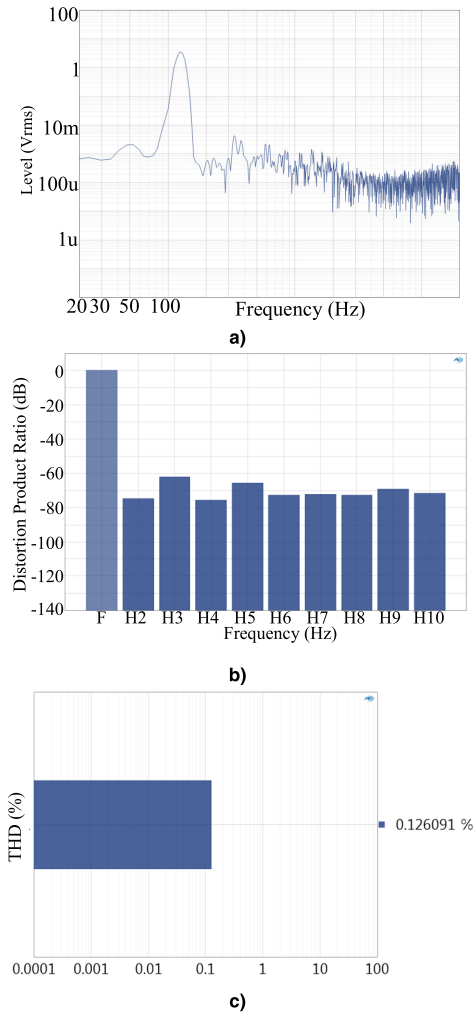


FIGURE 22. Harmonic analysis of RPDBSC load current considering back EMF (1250r/min). a) FFT. b) Harmonic distribution. c) THD.

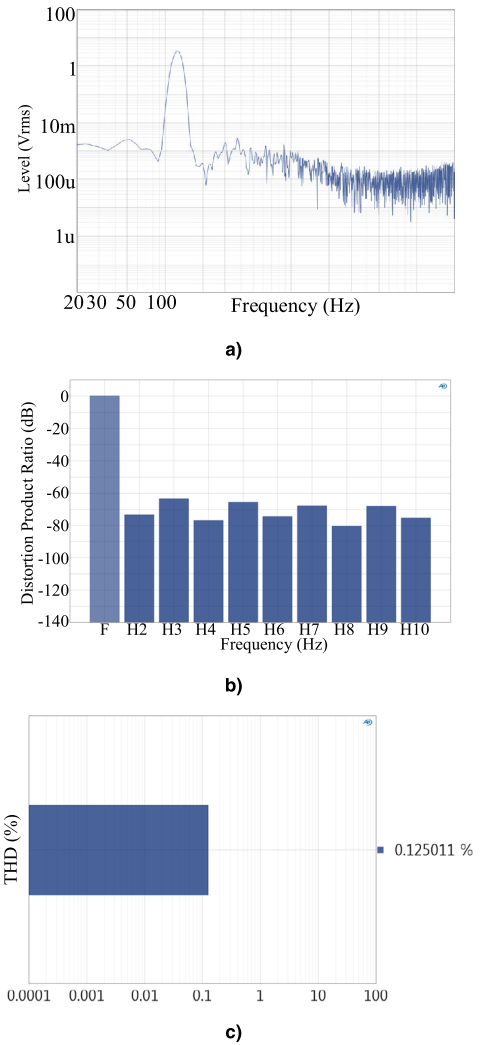


FIGURE 23. Harmonic analysis of RPDBSC load current considering back EMF (500r/min). a) FFT. b) Harmonic distribution. c) THD.

circuit. The soft-switching converter topology reshapes and optimizes the voltage waveform of the switching devices, as shown in Fig. 19a) and Fig. 19b), reduces the EMI of the system, which makes the switching devices operate in the state of soft-switching, and expands the switching frequency to 100 kHz. By analyzing the waveform in Fig. 19c), it can be seen that under the maximum output current, the current stress of the auxiliary resonant circuit is less than 22 A, and the voltage stress of the switching device is basically consistent with the bus voltage. The transient current waveforms are shown in Fig. 19d), and the transient adjustment process of current is relatively short due to high bandwidth (>5 kHz).

Fig. 20a) shows the amplitude and phase frequency characteristics of RPDBSC under the condition of underdamping. Adding the damping in the form of harmonic current feedback can effectively suppress the resonance peak (at 14.59 kHz) and improve the stability of the system. According to Fig. 20b), the closed-loop bandwidth of the system is more than 5 kHz and the phase margin is more than 60° .

Fig. 21b) shows the voltage waveform of the back EMF of the load motor at 1250 r/min. According to Fig. 21a), the total harmonic distortion (THD) of the corresponding motor back EMF is 2.45%. It can be seen from Fig. 21a) that the THD of output current is less than 0.2% under the condition of given load current of 12.5A and different back EMF (different motor speed) of load motor. Therefore, the closed-loop current control strategy can effectively restrain the influence of back EMF disturbance on current accuracy. Fig. 22a), 23a), 22b), 23b), 22c), and 23c) show the FFT of load current, harmonic content of each order and THD analysis results of load motor at 1250 r/min and 500 r/min respectively. The results show that the harmonic content A_H ($H = 2 \dots 10$) of the output current is less than -60 dB at 1250r/min and 500 r/min.

VI. CONCLUSION

In this paper, the current closed-loop control strategy of RPDBSC is researched. According to the closed-loop control

model, the harmonic current feedback active damping compensation is analyzed. By designing and optimizing the harmonic current feedback loop, the stability of the system is improved and the resonance peak is effectively suppressed. In addition, the influence of the back EMF on the current accuracy is analyzed, and a compensation method based on feedforward control is proposed. The conclusion can be summarized as follows.

(1) The harmonic current feedback can increase the damping of the system, which make the pole position of the closed-loop transfer function shift and move to the left half plane. It can improve the stability of the system and weaken the resonance peak.

(2) The back EMF affects the accuracy of the output current. Compared to the back EMF with harmonics, the back EMF without harmonics has less influence on the current accuracy.

(3) The influence of back EMF on current accuracy can be compensated by feedforward control. When the motor speed ranges from 250r/min to 1250r/min, high precision output current is realized.

Despite the analysis and control strategy optimization are based on RPDBSC, the proposed control method can also be applied to dual buck based converter.

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JIAN WEI received the B.E. degree from the Harbin Institute of Technology (HIT), China, in 2013, and the M.E. degree from the Beijing Institute of Space Launch Technology, China, in 2016. He is currently pursuing the Ph.D. degree with the Department of Electrical Engineering, HIT.

His current research interests include soft-switching converters and high-precision motor drive.



HAILIN ZHANG (Member, IEEE) was born in Chongqing, China. He received the B.E., M.E., and Ph.D. degrees in electrical engineering from the Harbin Institute of Technology, Harbin, China, in 2010, 2012, and 2017, respectively. He is currently a Postdoctoral Researcher with the Postdoctoral Station of Electrical Engineering, Chongqing University. His research interests include soft-switching converters and high-precision motor drive.



BAOQUAN KOU (Member, IEEE) received the B.E. and D.E. degrees from the Harbin Institute of Technology, China, in 1992 and 2004, respectively, and the M.E. degree from the Chiba Institute of Technology, Japan, in 1995.

From 2005 to 2007, he worked as a Postdoctoral Researcher at the Mobile Station, HIT. Since 2007, he has been a Professor with the School of Electrical Engineering and Automation, HIT. His research interests include electric drive of electric

vehicles, linear motors and linear electromagnetic drives, control of the power quality, and superconducting motors.



LU ZHANG (Member, IEEE) received the B.E., M.E., and D.E. degrees from the Harbin Institute of Technology (HIT), China, in 2008, 2010, and 2014, respectively.

Since 2017, he has been an Associate Professor with the School of Electrical Engineering and Automation, HIT. His research interests include magnetic levitation planar actuators and linear motors.



DEXUE SONG received the B.E. and M.E. degrees from the Shenyang University of Technology, China, in 2015 and 2018, respectively. She is currently pursuing the Ph.D. degree with the Department of Electrical Engineering, HIT. Her current research interest includes weakening of the axial flux permanent magnet synchronous motor.

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